

Power for One-Sample t-Test

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Introduction

Power analysis for the one-sample t -test determines the sample size required to detect a specified departure of the population mean from a reference value with a given probability, or equivalently the power achievable at a fixed sample size. The one-sample t -test arises in many practical contexts: validating a quality-control measurement against a target, comparing a single-arm pilot study against a historical benchmark, testing whether a within-subject change differs from zero. In every case, the planning question is the same — given the smallest effect of scientific interest and the assumed within-population standard deviation, how many observations are needed for an adequately-powered test? Pre-specifying the calculation in the protocol is now standard for any prospective study.

Prerequisites

A working understanding of the one-sample t -test, the concept of statistical power and type-II error, and Cohen's d as the standardised effect-size measure for one-mean tests.

Theory

Under the alternative hypothesis $H_1 : \mu = \mu_1 \neq \mu_0$, the t -statistic follows a non-central t -distribution with $n - 1$ degrees of freedom and non-centrality parameter

$$\lambda = \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}} = d\sqrt{n},$$

where $d = (\mu_1 - \mu_0)/\sigma$ is Cohen's standardised effect. Power is the probability $P(|T| > t_{1-\alpha/2, n-1})$ computed under the non-central t -distribution. Closed-form sample-size solutions invert this expression to produce the smallest n achieving target power against a specified d , α , and one-sided / two-sided test direction.

Assumptions

The population is approximately Normal or the sample size is large enough for the central limit theorem to apply, the within-population standard deviation σ is reasonably known (or has been estimated from pilot data with adequate precision), and the effect size d is pre-specified — usually as the smallest clinically or scientifically meaningful difference rather than the expected value.

R Implementation

```
library(pwr)

pwr.t.test(d = 0.4, sig.level = 0.05, power = 0.80, type = "one.sample")

pwr.t.test(n = 25, sig.level = 0.05, power = 0.80, type = "one.sample")

n_grid <- seq(10, 100, by = 5)
pw <- sapply(n_grid, function(n)
```

```
pwr.t.test(n = n, d = 0.4, type = "one.sample")$power)
plot(n_grid, pw, type = "l", lwd = 2, col = "#2A9D8F",
     xlab = "n", ylab = "Power", main = "One-sample t-test, d = 0.4")
abline(h = 0.80, lty = 2)
```

Output & Results

`pwr.t.test()` returns the sample size required to achieve target power against a specified d , the achievable power for a given n , or the minimum detectable effect size for a given n and target power. Plotting the power curve against n for the chosen d visualises the trade-off and is a useful supplement to the single-point calculation.

Interpretation

A reporting sentence: “A sample of 51 participants is required to detect a standardised effect of $d = 0.4$ with 80 % power at $\alpha = 0.05$ (two-sided) using a one-sample t -test against the reference value 10. With the planned sample of 25 participants, the minimum detectable effect at 80 % power would be $d = 0.58$, which exceeds the smallest clinically meaningful effect of 0.4 and would leave the study underpowered. The protocol therefore stipulates 51 participants, with 60 enrolled to allow for 15 % attrition.” Always state d , α , power, and the attrition margin.

Practical Tips

- The one-sample t -test power formula applies equally to the paired t -test if the effect size d is computed using the within-subject difference SD rather than the between-subject SD; this is the most common practical use of the calculation.
- For very small effect sizes, sample sizes grow rapidly: halving d quadruples the required n because power scales with $d\sqrt{n}$. A clear-headed choice of the smallest clinically meaningful d is therefore essential, and pilot-data d estimates should be treated with appropriate uncertainty.
- One-sided power at α approximately equals two-sided power at 2α when the true effect is in the hypothesised direction; the rule of thumb is useful for quick comparisons but exact non-central t computation is preferred for the protocol calculation.
- Choose d on the basis of the minimum clinically important difference (MCID) or smallest scientifically meaningful effect, not the expected effect from pilot data; powering for the expected effect leads to under-powered trials when the truth is more modest.
- Inflate the calculated sample size for expected attrition, dropout, or non-evaluable observations; a 15–20 % inflation is typical for clinical studies, with field-specific norms varying.
- Distinguish achievable power (post-hoc, computed from observed effect size) from planned power (a priori); only planned power is methodologically valid, and “post-hoc power” calculations are widely discouraged because they are mathematically determined by the observed p -value.

R Packages Used

`pwr::pwr.t.test()` for the canonical one-sample and paired t -test power calculation; `WebPower::wp.t()` for an alternative interface mirroring popular online calculators; `Superpower` for ANOVA and factorial-design power simulation; `simr` for power analysis via simulation when classical formulas do not apply; `MESS::power_t_test()` as a fast alternative.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**

- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Sample size, power, and Quarto reporting
- Power: closed-form and simulation